

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/325880246>

The Effectiveness of Resisted Sled Training (RST) for Sprint Performance: A Systematic Review and Meta-analysis

Article in *Sports Medicine* · June 2018

DOI: 10.1007/s40279-018-0947-8

CITATIONS

0

READS

200

4 authors, including:



Pedro E. Alcaraz

Universidad Católica San Antonio de Murcia

103 PUBLICATIONS **599** CITATIONS

[SEE PROFILE](#)



Jorge Carlos-Vivas

Universidad Católica de Murcia (UCAM)

7 PUBLICATIONS **5** CITATIONS

[SEE PROFILE](#)



Oribe Bruno

Universidad Católica San Antonio de Murcia

2 PUBLICATIONS **0** CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Have been wondering about the use of Sled Towing or Pulling Training for Speed? Exciting news! [View project](#)



Acute and Chronic Effect of Sled Towing Training on Speed and Athletic Performance: A Systematic Review [View project](#)



The Effectiveness of Resisted Sled Training (RST) for Sprint Performance: A Systematic Review and Meta-analysis

Pedro E. Alcaraz^{1,2} · Jorge Carlos-Vivas¹ · Bruno O. Oponjuru¹ · Alejandro Martínez-Rodríguez³

© Springer International Publishing AG, part of Springer Nature 2018

Abstract

Background Sprinting is key in the development and final results of competitions in a range of sport disciplines, both individual (e.g., athletics) and team sports. Resisted sled training (RST) might provide an effective training method to improve sprinting, in both the acceleration and the maximum-velocity phases. However, substantial discrepancies exist in the literature regarding the influence of training status and sled load prescription in relation to the specific components of sprint performance to be developed and the phase of sprint.

Objectives Our objectives were to review the state of the current literature on intervention studies that have analyzed the effects of RST on sprint performance in both the acceleration and the maximum-velocity phases in healthy athletes and to establish which RST load characteristics produce the largest improvements in sprint performance.

Methods We performed a literature search in PubMed, SPORTDiscus, and Web of Science up to and including 9 January 2018. Peer-reviewed studies were included if they met all the following eligibility criteria: (1) published in a scientific journal; (2) original experimental and longitudinal study; (3) participants were at least recreationally active and towed or pulled the sled while running at maximum intensity; (4) RST was one of the main training methods used; (5) studies identified the load of the sled, distance covered, and sprint time and/or sprint velocity for both baseline and post-training results; (6) sprint performance was measured using timing gates, radar gun, or stopwatch; (7) published in the English language; and (8) had a quality assessment score > 6 points.

Results A total of 2376 articles were found. After filtering procedures, only 13 studies were included in this meta-analysis. In the included studies, 32 RST groups and 15 control groups were analyzed for sprint time in the different phases and full sprint. Significant improvements were found between baseline and post-training in sprint performance in the acceleration phase (effect size [ES] 0.61; $p=0.0001$; standardized mean difference [SMD] 0.57; 95% confidence interval [CI] -0.85 to -0.28) and full sprint (ES 0.36; $p=0.009$; SMD 0.38; 95% CI -0.67 to -0.10). However, non-significant improvements were observed between pre- and post-test in sprint time in the maximum-velocity phase (ES 0.27; $p=0.25$; SMD 0.18; 95% CI -0.49 to 0.13). Furthermore, studies that included a control group found a non-significant improvement in participants in the RST group compared with the control group, independent of the analyzed phase.

Conclusions RST is an effective method to improve sprint performance, specifically in the early acceleration phase. However, it cannot be said that this method is more effective than the same training without overload. The effect of RST is greatest in recreationally active or trained men who practice team sports such as football or rugby. Moreover, the intensity (load) is not a determinant of sprint performance improvement, but the recommended volume is > 160 m per session, and approximately 2680 m per week, with a training frequency of two to three times per week, for at least 6 weeks. Finally, rigid surfaces appear to enhance the effect of RST on sprint performance.

✉ Pedro E. Alcaraz
pedro.e.alcaraz@gmail.com

✉ Alejandro Martínez-Rodríguez
amartinezrodriguez@ua.es

Extended author information available on the last page of the article

Key Points

Resisted sled training (RST) is an effective training method for the development of sprint performance, specifically in the early acceleration phase (≤ 10 m), independent of participant and load characteristics. However, the effects are larger in male, trained, and team sports athletes. Conversely, this method is not more effective than the unresisted sprint in improving sprint performance.

RST has a small effect on performance in the maximum-velocity phase (≥ 15 -m flying sprints at maximum intensity using a run-in distance of ≥ 10 m) and/or when performing sprints ≥ 20 m.

There is no optimal load for RST, and the load should be adapted according to the desired objective. However, when replicating the demands of sprinting (i.e., the movement pattern, load, and movement velocity) using sled towing with a slight overload, the load must never be $> 20\%$ of body mass.

1 Introduction

Sprinting is recognized as the fastest mode of unaided human locomotion. It is an important action that humans have employed since prehistoric times, not only as a determinant of survival but also as a key activity in the development and final results of competitions in a range of sport disciplines, both individual (e.g., athletics) and team sports (e.g., soccer, rugby, American football, basketball, futsal, or field hockey). For example, straight sprinting is the most frequent action in goal situations in professional soccer [1]. In addition, professional soccer players, for whom sprinting is key, have become faster over time [2, 3], independent of sex and/or age [4]. However, the maximum expression of a sprint is represented in the 100-m dash final in the Olympic games, a sporting event with some of the largest social media and sporting repercussions worldwide. Therefore, sprinting has been extensively studied from both a biomechanical and a physiologic point of view [5].

In a 100-m race, for instance, as in many other sports in which speed is of crucial importance, there are several clearly defined phases, but these can be summarized in two: acceleration and maximum-velocity phases. The acceleration phase is characterized by the athlete starting the sprint from a semi-static position and increasing their speed rapidly in a short period of time; in the maximum-velocity phase, the athlete moves at high speed that is maintained without any apparent acceleration. It is of utmost importance to emphasize that, in some cases, these phases are not related,

since different factors affect performance in each phase; in addition, not all sports include both phases [6]. Therefore, training protocols to develop each of these phases must also differ. For example, the main muscle groups involved in the acceleration phase are the ankle, knee, and hip extensors [7]. Given the limited time available to produce force during a sprint, the main manifestation of force is explosive, defined as the development of maximal force in minimal time, or rate of force development (RFD) [8]. An athlete's position, with a slight body lean, relates to the horizontal net ground reaction force (GRF), which is paramount to accelerate the body forward [9], with higher contact times (CTs) [9, 10] and ratio of forces (horizontal:vertical GRF), when compared with the maximum-velocity phase [9]. In contrast, the main muscle groups involved in the maximum-velocity phase are the hip and ankle extensors, which contribute to greater vertical GRF (vGRF) [11]. Weyand and colleagues [12, 13] stated that top speed was mechanically limited by maximal vGRF, presenting a strong positive relationship between top speed and the average vGRF applied during the first half of the stance period [14]. Therefore, it seems that, together with RFD, maximum (relative) strength may play an important role in this phase. It should be noted that the maximal-velocity phase includes an asymmetrical production of force and the RFD is very high [15] as a result of much shorter ground CT than observed in the acceleration phase [9, 16, 17]. In addition, the erect stance-phase posture that sprinters adopt in this phase likely contributes to the stiffness required to decelerate the limb and body relatively quickly after the instant of foot-ground impact [14], all in order to produce the maximum possible mechanical power that generates a high horizontal velocity. Notably, neuromuscular function is vital to sprint performance, because the activity and the interaction of the central nervous system with the muscles ultimately influence muscle RFD, given that the stretch-shortening cycle (SSC) actions are particularly prevalent in sports involving sprinting [15]. Thus, efficient usage of the SSC can recover approximately 60% of the total mechanical energy, with a higher contribution of non-metabolic energy sources with increases in running speed [18, 19].

For sprint training, given the above-mentioned factors, it is typical to recommend different classifications of training methods, such as that based on the specificity of the method and its similitude with the sprint's technique [20]. MacDougall and Sale [21] suggested that training should be specific with regards to movement pattern, contraction velocity, muscle activation type, and applied force. In fact, a recent study showed that chronic performance and biomechanical adaptations are associated with vertically and horizontally oriented conditioning regimens [22]. Plisk [20] established primary methods that simulate the sprint movement pattern (sprint-technique drills,

stride length and frequency exercises, and sprints of varying distances and intensities); secondary methods that simulate the sprint action but with a slight overload or degree of assistance (resisted or assisted methods, respectively); and tertiary methods, characterized by non-specific sprint development (resistance training, plyometric training, complex training, stretching, etc.) [23–25]. It is also typical to classify sprint training methods according to the force–velocity (F–V) relationship of muscle shortening, described by Hill and colleagues [26, 27] many years ago. The F–V relationship describes a characteristic property of the muscle that dictates its power-production capacities [28]. Because mechanical power is the expression of both force and velocity, it is generally accepted that improving force-production potential and/or velocity of movement is effective when seeking to improve short-distance sprinting [29]. Based on contemporary scientific knowledge, it is evident that maximal (relative) strength, the RFD, and peak power-generating capacity are all important attributes that need to be developed when implementing strength and conditioning programs [30] to increase sprint performance during the entire season. Haff and Nimphius [30] suggested a mixed-methods approach in which a variety of loads and exercise types are used in a periodized fashion to optimize power output. In this context, resisted and assisted sprint training protocols play a key role [30].

One of the most traditional secondary methods to improve sprint performance is resisted sled training (RST), mainly due to its greater effect on horizontal forces [31, 32] when compared with tertiary methods, which have a greater vertical orientation of resistance forces [29]. RST has been applied in sports for decades, evolving from the pull of a wheel to current electromechanical devices that regulate the load to produce the loss of speed desired by the athlete (i.e., 1080 Motion™). Traditionally, and mainly addressing the principle of specificity, the most popular manuals and training guides [33, 34] have recommended that normal unresisted (UR) sprint biomechanics should be maintained when RST is used. Hence, loads should be chosen based on the sport and the athlete's physical status. For example, track and field sprinters may use loads that do not decrease running velocity by more than 10–12% of body mass (BM) [35–37]. In contrast, field sport athletes who overcome external resistance while blocking and tackling can use loads 20–30% of BM to improve early acceleration [38, 39]. This hypothesis has been widely accepted, since it is believed that the production of mechanical power when the load increases is significantly reduced [40]. This reduction of mechanical power is associated with an improper configuration of the athlete's levers (involving different motor units or even muscle groups), an increase in the CT, and no possibility of adequate use of the SSC. Thus, recent studies have tried to elucidate the load that maximizes mechanical power in

RST, with some resulting controversy. Monte et al. [40] indicated that the maximum power produced when using RST occurs with loads close to 20% BM, without inducing significant changes in the sprint technique when this load was used, whereas Cross et al. [41] indicated that the maximum power output is achieved with loads near 80% BM. However, this latter study [41] has the limitation of not having measured the effect on kinematic parameters and measuring the horizontal power when the athletes achieved the maximum velocity of sprint, given that the maximum power output in sprinting is known to usually occur in the first steps. Moreover, the horizontal force and power decreased by 82.0% and 62.5%, respectively, from the first to the last step during the 20-m sprint with sled towing [40]. Interestingly, both studies have used the method developed by Samozino et al. [42] to calculate the horizontal power production but in different disciplines (sprinters vs. soccer players) and in different phases. These authors indirectly estimated the anterior-posterior (horizontal) force of a sprint from spatio-temporal parameters and, consequently, calculated the horizontal power production during the sprint.

From a kinematic point of view, cross-sectional studies have focused on the effects of RST on different variables such as stride length (SL), CT, flight times (FT), and joint angles. For example, Alcaraz et al. [31] showed a decrease in SL and running velocity with an RST load of 16% BM but observed no significant changes in running technique (i.e., by analyzing the joint angles). In addition, various studies [37, 40, 43, 44] showed a decrement in FT and SL, and an increase in CT as a function of sled load. From a kinetic point of view, Martínez-Valencia et al. [45] found an acute, significant RFD increase when added loads were 15–20% BM compared with an UR sprint. In addition, other authors have concluded that RST, mainly with high loads (30% BM), increases relative net horizontal and propulsive impulse production compared with UR sprinting by directing force production more horizontally and by allowing longer time to apply force against the ground [32], which could be transferred to a better application of horizontal GRF. However, the same research team compared the short-term effect (8 weeks) of using RST with low loads (~13% BM) versus higher loads (~43% BM) [39] and did not find significant differences between the groups, either for improvement of performance or horizontal GRFs. Therefore, there is no agreement on whether low or high loads are more beneficial from both a kinetic and a kinematic point of view when increases in performance in the short and medium term are sought.

In sprinting with RST, the additional resistance experienced by the athlete arises mainly from the frictional force between the base of the sled and the running surface. This frictional force is approximately proportional to the total weight of the sled, and the coach may manipulate the

resistance experienced by the athlete by changing the weight placed on the sled. The greater the added weight, the greater the friction, and hence the slower the athlete's acceleration and the lower the maximum speed achieved by the athlete. The coefficient of friction between the sled and the running surface is determined by the type and surface roughness of the materials used in the base of the sled and running surface. For this reason, the use of different surfaces (grass vs. athletics track) can produce different stimuli with the same load [46, 47]. Another factor that can alter friction is the inertia of the sled, which is higher when the sled moves at low speed.

Finally, a recent systematic review by Petrakos et al. [47] described in detail the training recommendations in the existing literature. However, the authors concluded that performance benefits of RST over UR training remain to be conclusively demonstrated. Between-study comparisons are limited, primarily due to discrepancies in participant training status and level and sled load prescription, indicating that future work is required to define the optimal training load for RST, depending on the specific components of sprint performance to be enhanced.

In an effort to clarify discrepancies in the literature, the following questions should be addressed: (1) does RST improve sprint performance; (2) will the effect be greater when RST is used for the different phases of sprint; (3) is RST more effective than UR sprinting; (4) what is the optimal load when applying RST for obtaining higher sprint adaptations; (5) should the load be different for the different phases of sprint; (6) does RST have a different effect on athletes depending on age; (7) will the effect be greater if the athlete is highly trained; (8) how many days per week should RST be applied; (9) for how many weeks should RST be applied; and (10) can the surface affect the performance adaptations?

Therefore, and in response to the questions raised by the current literature, the objective of this systematic review with meta-analysis is twofold: (1) to review the state of the current literature on intervention studies that have analyzed the effects of RST on sprint performance both in the acceleration and the maximum-velocity phase in healthy athletes and (2) to establish which RST load characteristics are associated with the greatest improvements in sprint performance.

2 Methods

We followed the Preferred Reporting Items for Systematic review and Meta-Analyses (PRISMA) guidelines [48] for search procedures, study selection, data collection, and analysis.

2.1 Literature Research and Data Sources

The search was performed by two independent reviewers (AM and BO) using the following databases to identify studies for this review: PubMed, SPORTDiscus, and Web of Science. The search results were limited to studies published up to and including 9 January 2018. Reviewers used a computerized and manual library search with the following Boolean search phrases in all of the mentioned databases: ("Sprint" OR "Pulling" OR "Towing" OR "Training") AND ("Sled" OR "Resisted"). Figure 1 shows the flow diagrams for the entire search process for both variables.

2.2 Inclusion Criteria

The following inclusion criteria had to be met for a study to be considered for this review: (1) type of publication: the current research only considered articles published in scientific journals; (2) type of study: original experimental and longitudinal studies were considered; (3) participants: the study participants were at least recreationally active who towed or pulled the sled while running at maximum intensity; (4) intervention: sled towing and sprinting must have been one of the training methods used; (5) outcome measures: studies must have identified the load of the sled used, distance tested up to a maximum of 100 m in at least one of sprint time and/or sprint velocity, pre-test and post-test results, and test used; (6) tests must have been measured by an automated electronic machine, such as timing gates or a radar gun, or by a manual method such as a stopwatch; (7) only studies in English were considered; (8) quality assessment: a score > 6 points on the Physiotherapy Evidence Database scale (PEDro) for systematic review [49–51].

All studies that included the time used to cover a maximum distance of 10 m were utilized for the subgroup analysis of the acceleration phase. All studies that measured the time used to cover a distance of at least 15 m at maximum intensity using a run-in distance of ≥ 10 m before recording time were used for the subgroup analysis of the maximum-velocity phase. All studies that included the time used to cover a distance of at least 20 m were used for the subgroup analysis of the full sprint. For studies that did not include the time but showed the average speed and the covered distance, the time was calculated as $\text{time} = \text{distance} / \text{average speed}$. Likewise, for the studies that included the sprint time and presented the acceleration phase and the full sprint data, or the acceleration phase and the maximum-velocity phase, the time was calculated as the subtraction or summation (based on the conditions, respectively) of the known mean data.

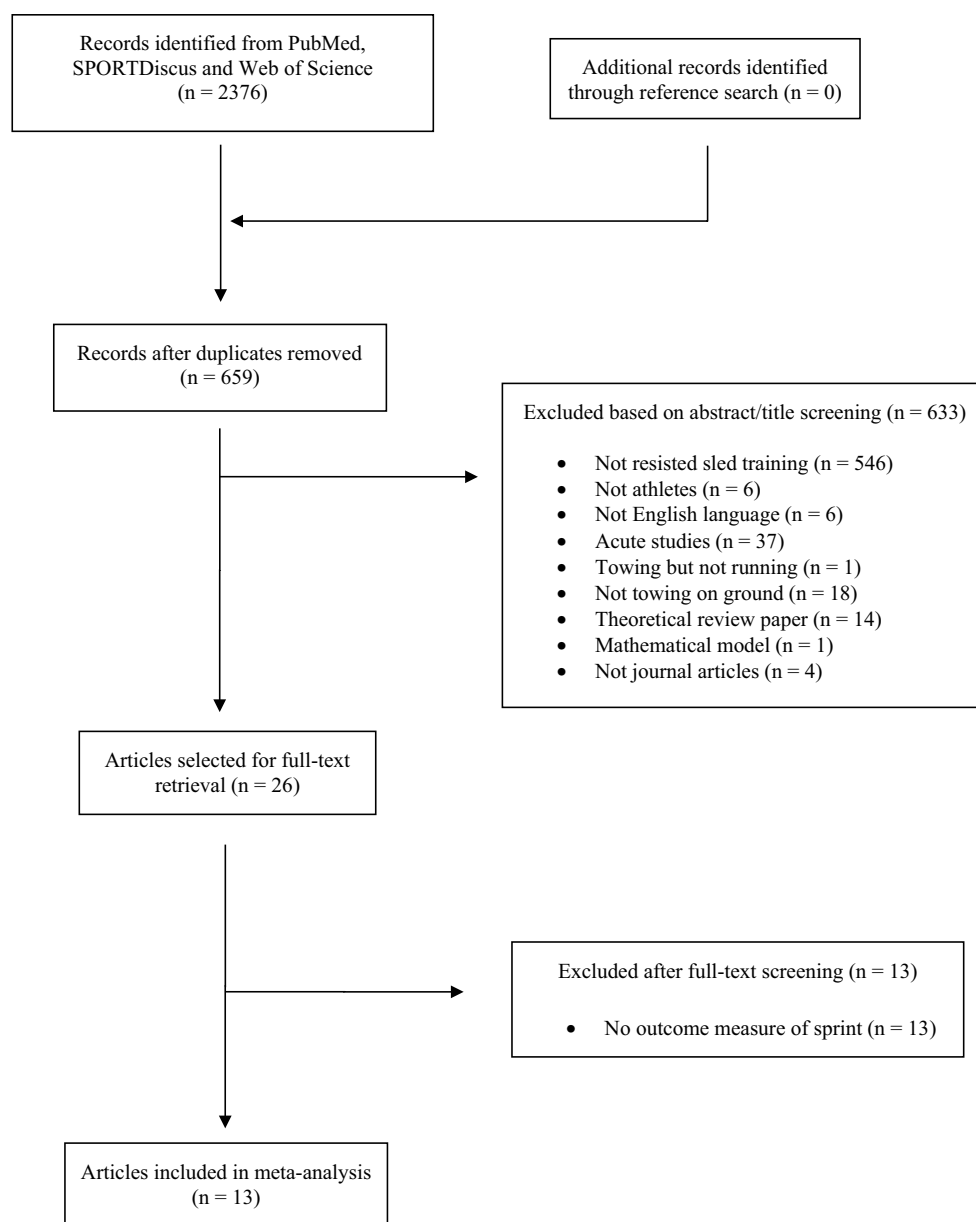


Fig. 1 Flow diagram of the process of study selection

2.3 Study Selection

Two reviewers (AM and BO) independently evaluated the titles and abstracts of the studies that resulted from the search. Disagreements between the two reviewers were resolved by discussion; if necessary, a third reviewer (PA) was consulted to reach a consensus.

For this review, only studies that met the eligibility criteria were selected. Reviewers independently assessed the methodological quality of the eligible studies using the PEDro scale [49–51].

2.4 Data Extraction and Analysis

One reviewer (AM) extracted the following information from each full-text article, and a second reviewer (BO) confirmed the extraction. Disparities in data abstraction were resolved by a third reviewer (PA).

Review Manager Software (RevMan 5.2; Cochrane Collaboration, Oxford, UK) and Comprehensive Meta-analysis software (Version 2; Biostat, Englewood, NJ, USA) were used for meta-analysis. A randomized effect model was used. Heterogeneity among studies was assessed using I^2 statistics.

Subgroup analyses were performed to evaluate the potential moderating factors or variables. For continuous variables comparison, the cut-off values based on medians from the full sprint analysis were used. However, in specific cases, the cut-off was established in an arbitrary way (i.e., load). Publication bias was evaluated using the estimating funnel plot asymmetry test. A p value of <0.05 was considered statistically significant. The standard deviation (SD) was calculated as the square root of the summation of the squared SDs of the mean time in the known conditions. Cohen's d was used to calculate the effect size (ES, 95% confidence limit) of each study using the following equation [52]:

$$ES = \frac{M_{pre} - M_{post}}{Spre} \left(1 - \frac{3}{4n - 5} \right)$$

where M_{pre} is the mean value before the CT intervention, M_{post} is the mean value after the intervention, n is the sample size of the CT group, and $Spre$ is the SD pre-intervention.

Threshold values for Cohen's ES statistics were >0.2 (small), >0.6 (moderate), and >1.2 (large) [53].

2.5 Risk of Bias

Methodological quality and risk of bias were independently assessed via visual interpretation of the funnel plot by two authors (AM, BO), with disagreements resolved by a third party evaluator (PA), in accordance with Cochrane Collaboration guidelines [54].

3 Results

3.1 Characteristics of Included Studies

A total of 2376 studies were found following the study selection procedures, and 659 studies remained after duplicates were removed. Finally, 13 studies [39, 55–66] were included in this meta-analysis (Table 1). All included studies had an RST group that accounted for a total of 32 subgroups analyzed for sprint time. However, only ten studies [55–62, 64, 66] had control groups, which represented a total of 15 subgroups analyzed.

The quality (internal validity) of the trials, according to a PEDro scale [51], was high. The mean score was eight of a possible ten points.

3.2 Characteristics of the Interventions

Table 2 shows the characteristics of the different RST intervention groups. The sled-towing exercise load performed ranged from 5 to 80% of BM. The interventions ranged from 4 to 10 weeks in duration, with a frequency of one to three

sessions-week⁻¹. The distance covered in sprint assessment ranged from 10 to 50 m. Regarding the sprint time assessment, nine studies used photoelectric cells [39, 55, 57, 58, 61–65], three studies used a stopwatch [56, 59, 60], and only one study used an indirect method [66] that was recently validated by Samozino et al. [42] to record sprint performance.

3.3 Acceleration Phase

The effect of RST on sprint time was measured in 144 participants. The results of the overall effects on sprint time showed a significant and moderate improvement between pre- and post-test on the sprint performance (ES 0.61; $p=0.0001$; standardized mean difference [SMD] 0.57; 95% confidence interval [CI] -0.85 to -0.28), with an average heterogeneity of $I^2=28\%$ (Fig. 2) [39, 55–57, 59–66]. Furthermore, in the studies that included a control group, a non-significant improvement was found in participants belonging to the RST group compared with the control group (ES 0.09; $p=0.64$; SMD 0.07; 95% CI -0.37 to 0.23), with an average heterogeneity of $I^2=0\%$ (Fig. 3) [55–57, 59–62, 64, 66].

Table 3 presents the subgroup analysis assessing potential moderating factors for sprint time on the acceleration phase of sprint. Regarding the population characteristics, significant ($p \leq 0.05$) improvements were found for age, sex, and level. Moderate ESs were obtained for age (ES $<21=0.60$; $\geq 21=0.62$), male sex (ES 0.73), recreationally active (ES 0.75), and trained (ES 0.84). However, non-significant improvements were found for female sex (ES 0.14) and highly trained (ES 0.30). Additionally, between-subgroup analyses revealed significant ($p \leq 0.05$) differences for sex and level.

Concerning the exercise characteristics, significant ($p \leq 0.05$) improvements were found for lower loads ($<20\%$ BM) (Fig. 4), training frequency, training period duration, session volume, total training volume, and rigid and grass surfaces. A large ES was obtained for a frequency over twice a week (ES 1.85) [55], and moderate ESs were found for lower loads (ES 0.61) [39, 55–57, 59–65], >6 -week training periods (ES 0.63) [39, 55, 56, 60, 63–66], session volume >160 m (ES 0.92) [55, 56, 59, 60, 64], total weekly training volume >2680 m (ES 0.83) [55, 59, 60, 64], and rigid surface (ES 0.69) [39, 56, 61, 64]. Small ESs were also found for a training frequency equal to or fewer than two trainings per week (ES 0.52) [39, 56, 57, 59–66], a total weekly training volume <2680 m (ES 0.53) [39, 56, 57, 61–63, 65, 66], and grass surface (ES 0.47) [59, 60, 65, 66]. However, non-significant improvements were found for higher loads ($\geq 20\%$ BM; ES 0.63) [39, 63, 66], a training period ≤ 6 weeks (ES 0.55) [57, 59, 61, 62], and track surface (ES 0.64) [55, 57, 62, 63]. Additionally, significant ($p \leq 0.05$) differences were found for frequency of training between subgroups; however, it should be taken into

Table 1 Main characteristics^a of studies included in the meta-analysis

Study	Type	N		Females (%)	Age (years)	Weight	Height	Level
		CG	RST					
Alcaraz et al. [62]	A: 0–15 m B: 15–50 m C: 0–50 m	11	11	45	21.5 ± 2.2	69.8 ± 14.7	173.0 ± 10.5	Highly trained
Bachero-Mena et al. [63]	LL–A: 0–20 m	–	7	0	21.9 ± 2.3	75.8 ± 10.7	180.9 ± 6.8	Recreationally active
	LL–B: 20–40 m							
	LL–C: 0–40 m							
	ML–A: 0–20 m	–	6	0	20.8 ± 2.1	66.8 ± 8.5	173.8 ± 4.6	
	ML–B: 20–40 m							
	ML–C: 0–40 m							
Bachero-Mena et al. [63]	HL–A: 0–20 m	–	6	0	19.8 ± 1.6	70.2 ± 11.9	175.4 ± 6.8	
	HL–B: 20–40 m							
	HL–C: 0–40 m							
Clark et al. [58]	B: 18.3–54.9 m	7	7	0	19.7 ± 1.0	87.9 ± 17.3	181.15 ± 6.8	Trained
De Hoyo et al. [65]	A: 0–20 m	–	13	0	17.0 ± 1.0	73.1 ± 2.56	178.24 ± 1.3	Highly trained
	B: 20–50 m							
	C: 0–50 m							
Harrison and Bourke [57]	A: 0–10 m	7	8	0	20.5 ± 2.8	87.0 ± 10.5	NA	Highly trained
Kawamori et al. [39]	LL: 0–10 m	–	11	0	22.3 ± 5.2	82.5 ± 9.0	183.0 ± 0.07	Trained and recreationally active
	HL: 0–10 m		10	0	22.8 ± 3.3	77.5 ± 7.3	179.0 ± 0.08	
Lockie et al. [59]	A: 0–10 m	9	9	0	23.1 ± 4.2	83.1 ± 8.6	182.0 ± 0.1	Trained
Luteberget et al. [64]	A: 0–10 m	8	10	100	20.4 ± 3.1	74.6 ± 5.9	170.3 ± 5.3	Highly trained
	B: 10–30 m							
	C: 0–30 m							
Makaruk et al. [60]	A: 0–20 m	12	12	100	22.0 ± 0.9	61.5 ± 4.7	167.0 ± 0.1	Recreationally active
Morin et al. [66]	A: 0–20 m	6	10	0	26.3 ± 4.0	74.5 ± 5.3	177.0 ± 0.1	Trained
Spinks et al. [56]	A: 0–15 m	10	10	0	21.8 ± 4.2	83.3 ± 8.7	181.9 ± 6.2	Trained
West et al. [61]	A: 0–10 m	10	10	0	26.8 ± 3.0	90.2 ± 10.3	186.0 ± 8.0	Highly trained
	B: 10–30 m							
	C: 0–30 m							
Zafeiridis et al. [55]	A: 0–20 m	11	11	0	20.1 ± 1.9	73.1 ± 2.4	178.0 ± 7.0	Recreationally active
	B: 20–50 m							
	C: 0–50 m							

Data are mean, mean ± standard deviation, or *n*

A acceleration phase distance, B maximum-velocity phase distance, C full sprint distance, CG control group, HL high load, LL low load, ML medium load, NA not available, RST resisted sled training exercise-group

^aAll characteristics refer to the RST group

consideration that only one study evaluated a training frequency of more than twice a week.

3.4 Maximum-Velocity Phase

The effect of RST on sprint time in the maximum-velocity phase was measured in 81 participants. The results of the overall effects on sprint time showed a non-significant improvement between pre- and post-test on the sprint time (ES 0.27; $p=0.25$; SMD 0.18; 95% CI –0.49 to 0.13), with an average heterogeneity of $I^2=0\%$ (Fig. 5) [55, 58, 61–65].

Furthermore, the studies with a control group found a non-significant improvement in the RST group compared with the control group (ES 0.29; $p=0.23$; SMD 0.26; 95% CI –0.16 to 0.68), with an average heterogeneity of $I^2=5\%$ (Fig. 6) [55, 58, 61, 62, 64].

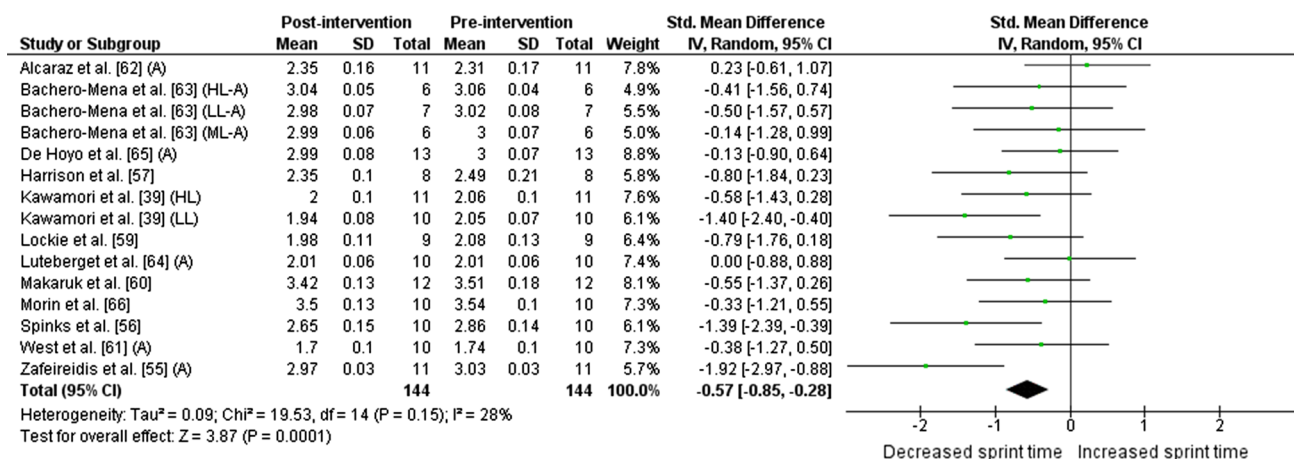
Subgroup analysis assessing potential moderating factors for sprint time on the maximum-velocity phase of sprint is presented in Table 4. Both for population and exercise characteristics, the ESs were small or trivial (ES 0.00–0.43) in the maximum-velocity phase (Fig. 7).

Table 2 Characteristics of the resisted sled training interventions and sprint time assessment of the studies included in the meta-analysis

Study	Fre- quency (week ⁻¹)	Session volume (m)	Total train- ing volume (m)	Duration (weeks)	Surface	Load (% BM)	Sprint time assessment	
							Instrument	Total distance (m)
Alcaraz et al. [62]	2	90–180	1080	4	Track	~8 to 9	Photoelectric cells	50
Bachero-Mena et al. [63]	2	100–210	2115	7	Track	5 12.5 20	Photoelectric cells	40
Clark et al. [58]	2	240–400	4060	7	Rigid	10.2	Photoelectric cells	36.6
De Hoyo et al. [65]	1–2	120–200	2680	8	Grass	12.6	Photoelectric cells	50
Harrison and Bourke [57]	2	120	1440	6	Track	~13	Photoelectric cells	10
Kawamori et al. [39]	2	90–140	1740	8	Rigid	~13 ~43	Photoelectric cells	10
Lockie et al. [59]	2	195–320	3100	6	Grass	12.6	Velocimeter with stopwatch	10
Luteberget et al. [64]	2	240–280	5200	10	Rigid	12.4	Photoelectric cells	30
Makaruk et al. [60]	2	180–360	6210	9	Grass	7.5–10	Stopwatch	20
Morin et al. [66]	2	100	1600	8	Grass	80	Indirect method	20
Spinks et al. [56]	2	215–340	4090	8	Rigid	12.6	Stopwatch	15
West et al. [61]	2	60	720	6	Rigid	12.6	Photoelectric cells	30
Zafeiridis et al. [55]	3	280	6720	8	Track	~6.8	Photoelectric cells	50

Data are mean or range

BM body mass

**Fig. 2** SMD between post and pre-intervention for sprint time in the acceleration phase. Squares represent the SMD for each trial. Diamonds represent the pooled SMD across trials. A acceleration phase,

CI confidence interval, HL high load, IV independent variable, LL low load, ML moderate load, SD standard deviation, SMD standardized mean difference

3.5 Full Sprint

The effect of RST on sprint time in the full sprint was measured in 96 participants. The results of the overall effects on sprint time showed a significant ($p \leq 0.05$) improvement between pre- and post-test in sprint performance (ES 0.36; $p = 0.009$; SMD 0.38; 95% CI -0.67 to -0.10), with an

average heterogeneity of $I^2 = 0\%$ (Fig. 8) [55, 60–66]. However, the studies with a control group found a non-significant improvement in the RST groups compared with the control groups (ES 0.05; $p = 0.89$; SMD 0.03; 95% CI -0.40 to 0.47), with an average heterogeneity of $I^2 = 0\%$ (Fig. 9) [55, 61, 62, 64].

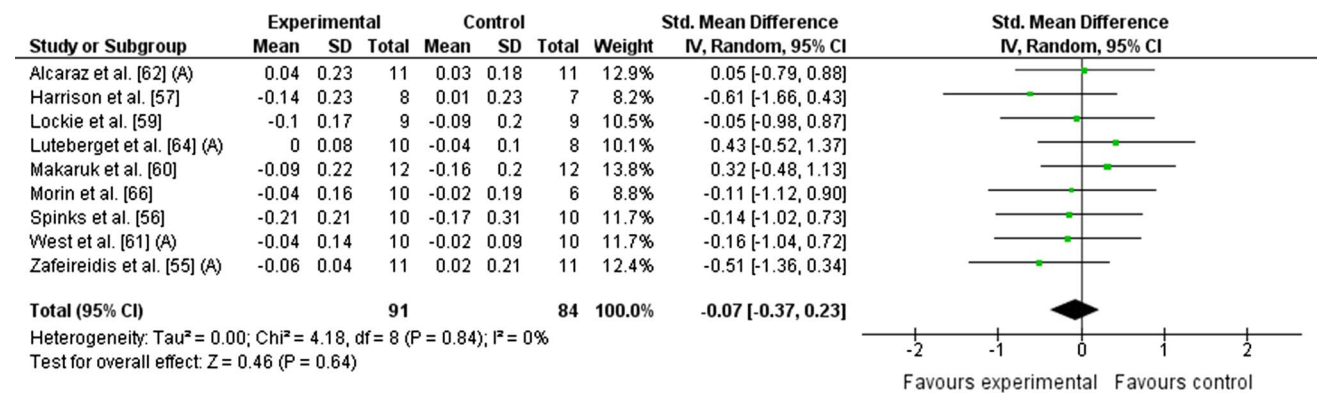


Fig. 3 SMD in post-intervention sprint time between intervention and control athletes for sprint time in the acceleration phase. Squares represent the SMD for each trial. Diamonds represent the pooled SMD

Table 5 presents the subgroup analysis assessing potential moderating factors for sprint time on the full sprint. Small ESs (0.24–0.53) were found for both population and exercise characteristics. Furthermore, significant ($p \leq 0.05$) improvements with small ESs were found for younger athletes (aged < 21 years; ES 0.37) [55, 63–65], male sex (ES 0.34) [55, 61, 63, 65, 66], team-sports athletes (ES 0.42) [60, 61, 64–66], using a load < 20% BM (ES 0.35) (Fig. 10) [55, 60–65], a training frequency equal to or less than twice a week (ES 0.35) [60–66], a period of training > 6 weeks (ES 0.39) [55, 60, 63–66], a session volume > 160 m (ES = 0.53) [55, 60, 64], and total training volume > 2680 m (ES 0.53) [55, 60, 64].

3.6 Evaluation of Potential Bias

Visual interpretation of the funnel plot was performed to evaluate potential bias. SMD between pre- and post-intervention sprint time in RST participants was considered notably symmetrical, suggesting the absence of a significant publication bias. Similar results were obtained for the evaluation of potential bias of the SMD in post-intervention sprint time between RST and control group athletes.

4 Discussion

RST has been used extensively to improve sprint performance. However, because there are many variables to manipulate (load, distance, friction, etc.) with this type of device, substantial controversy has been generated, both in the scientific community and in the field of training, regarding the optimal training protocol. The main objective of this systematic review with meta-analysis was to determine

whether RST effectively improves sprint performance, in both its acceleration and maximum-velocity phases. Second, we intended to establish which variables are associated with the largest RST-induced improvements. The main finding of the present analysis is that RST improves sprint performance, mainly in its early acceleration phase. However, its effect is trivial or low in the maximum-velocity phase or in sprints of ≥ 20 m. Also, no additional benefit was observed when RST was compared with UR conditions. Furthermore, it was determined that the magnitude of its effect on sprint performance is related to the selected population and/or training characteristics.

4.1 Acceleration Phase

As mentioned in Sect. 1, training with sled towing is an appropriate method for improving the early acceleration phase of the sprint (here defined as 0–10 m), with a moderate and significant effect. However, it is no more effective than performing the same sprint training without overload. The explanation for these results is that most studies use loads close to a subject's BM, making the differences in load very small. This fact has led some researchers [41, 66] to think that higher loads may have to be used for the improvements to be significantly greater than when lower loads are used. However, given the low number of studies using loads > 20% BM, the effect is similar (0.61 vs. 0.63, respectively), and the effect pre-post is not significant with high loads. This finding may be explained by some of the main characteristics that make the human being run at great speed, such as muscular mechanical properties (i.e., the capacity for adequate use of the elastic elements, both in series and in parallel), considering here the SSC, RFD, as well as the H reflex. It is clear that the SSC and H reflex can be developed in situations

Table 3 Subgroup analysis assessing potential moderating factors for sprint time in the acceleration phase in the studies included in the meta-analysis

Population characteristics	Studies		Resisted sled training				
	Number ^a	References	SMD (95% CI)	ES	<i>I</i> ² (%)	<i>p</i>	<i>p</i> _{Difference}
Age (~ 17 to 27 years)							
< 21 y.o.	7	Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Harrison and Bourke [57] Luteberget et al. [64]: A Zafeiridis et al. [55]: A	−0.53 (−1.01 to −0.04)	0.60	40	0.03	0.81
≥ 21 y.o.	8	Alcaraz et al. [62]: A Kawamori et al. [39]: LL, HL Lockie et al. [59] Makaruk et al. [60] Morin et al. [66] Spinks et al. [56] West et al. [61]: A	−0.60 (−0.97 to −0.23)	0.62	25	0.001	
Sex							
Female ≥ 45% sample	3	Alcaraz et al. [62]: A Luteberget et al. [64]: A Makaruk et al. [60]	−0.12 (−0.60 to 0.37)	0.14	0	0.64	0.05
Male	12	Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL, HL Lockie et al. [59] Morin et al. [66] Spinks et al. [56] West et al. [61]: A Zafeiridis et al. [55]: A	−0.70 (−1.01 to −0.39)	0.73	20	<0.001	
Level							
Recreationally active	7	Bachero-Mena et al. [63]: LL-A, ML-A, HL-A Kawamori et al. [39]: LL, HL Makaruk et al. [60] Zafeiridis et al. [55]: A	−0.79 (−1.24 to −0.34)	0.75	29	<0.001	0.04
Trained	5	Kawamori et al. [39]: LL, HL Lockie et al. [59] Morin et al. [66] Spinks et al. [56]	−0.85 (−1.27 to −0.43)	0.84	0	<0.001	
Highly trained	5	Alcaraz et al. [62]: A de Hoyo et al. [65]: A Harrison and Bourke [57] Luteberget et al. [64]: A West et al. [61]: A	−0.17 (−0.56 to 0.22)	0.30	0	0.39	
Sport characteristics							
Team sports	9	de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL, HL Lockie et al. [59] Luteberget et al. [64]: A Morin et al. [66] Spinks et al. [56] West et al. [61]: A	−0.58 (−0.91 to −0.26)	0.66	11	<0.001	0.07
Individual sports	1	Alcaraz et al. [62]: A	0.23 (−0.61 to 1.07)	0.23	NA	0.59	

Table 3 (continued)

Population characteristics	Studies		Resisted sled training				
	Number ^a	References	SMD (95% CI)	ES	<i>I</i> ² (%)	<i>p</i>	<i>p</i> _{Difference}
Exercise characteristics							
Load (5–80% BM)							
< 20% BM	12	Alcaraz et al. [62]: A Bachero-Mena et al. [63]: LL-A, ML-A de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL Lockie et al. [59] Luteberget et al. [64]: A Makaruk et al. [60] Spinks et al. [56] West et al. [61]: A Zafeiridis et al. [55]: A	−0.61 (−0.97 to −0.25)	0.61	43	<0.001	0.63
≥ 20% BM	3	Bachero-Mena et al. [63]: HL-A Kawamori et al. [39]: HL Morin et al. [66]	−0.45 (−0.99 to 0.10)	0.63	0	0.11	
Frequency (1–3 week^{−1})							
≤ 2 week ^{−1}	14	Alcaraz et al. [62]: A Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL, HL Lockie et al. [59] Luteberget et al. [64]: A Makaruk et al. [60] Morin et al. [66] Spinks et al. [56] West et al. [61]: A	−0.47 (−0.72 to −0.22)	0.52	0	<0.001	0.008
> 2 week ^{−1}	1	Zafeiridis et al. [55]: A	−1.92 (−2.97 to −0.88)	1.85	NA	<0.001	
Duration (4–10 weeks)							
≤ 6 weeks	4	Alcaraz et al. [62]: A Harrison and Bourke [57] Lockie et al. [59] West et al. [61]: A	−0.38 (−0.87 to 0.11)	0.55	12	0.13	0.41
> 6 weeks	11	Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Kawamori et al. [39]: LL, HL Luteberget et al. [64]: A Makaruk et al. [60] Morin et al. [66] Spinks et al. [56] Zafeiridis et al. [55]: A	−0.64 (−0.99 to −0.28)	0.63	35	<0.001	
Session volume (60–360 m)							
≤ 160 m	10	Alcaraz et al. [62]: A Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL, HL Morin et al. [66] West et al. [61]: A	−0.40 (−0.70 to −0.11)	0.46	0	0.008	0.18

Table 3 (continued)

Population characteristics	Studies		Resisted sled training				
	Number ^a	References	SMD (95% CI)	ES	<i>I</i> ² (%)	<i>p</i>	<i>p</i> _{Difference}
> 160 m	5	Lockie et al. [59] Luteberget et al. [64]: A Makaruk et al. [60] Spinks et al. [56] Zafeiridis et al. [55]: A	−0.89 (−1.53 to −0.25)	0.92	57		0.006
Total training volume (720–6720 m)							
≤2680 m	11	Alcaraz et al. [62]: A Bachero-Mena et al. [63]: LL-A, ML-A, HL-A de Hoyo et al. [65]: A Harrison and Bourke [57] Kawamori et al. [39]: LL, HL Morin et al. [66] Spinks et al. [56] West et al. [61]: A	−0.49 (−0.79 to −0.19)	0.53	9		0.001 0.48
> 2680 m	4	Lockie et al. [59] Luteberget et al. [64]: A Makaruk et al. [60] Zafeiridis et al. [55]: A	−0.78 (−1.52 to −0.03)	0.83	62		0.04
Surface							
Rigid	5	Kawamori et al. [39]: LL, HL Luteberget et al. [64]: A Spinks et al. [56] West et al. [61]: A	−0.71 (−1.24 to −0.18)	0.69	40		0.009 0.70
Track	6	Alcaraz et al. [62]: A Bachero-Mena et al. [63]: LL-A; ML-A; HL-A Harrison and Bourke [57] Zafeiridis et al. [55]: A	−0.57 (−1.19 to 0.05)	0.64	54		0.07
Grass	4	de Hoyo et al. [65]: A Lockie et al. [59] Makaruk et al. [60] Morin et al. [66]	−0.42 (−0.84 to 0.01)	0.47	0		0.05

Subgroup analyses were performed on SMD between post and pre-intervention sprint time. Median values of continuous variables were used as cut-off values for grouping studies. Changes in moderating factors were calculated as post-intervention minus pre-intervention values

A acceleration phase, B maximum-velocity phase, BM body mass, C full sprint, CI confidence interval, ES Cohen's *d* effect size, HL high load, *I*² heterogeneity, LL low load, ML medium load, NA not available, *p* test for overall effect, *p*_{Difference} test for subgroup differences, SMD standardized mean difference

^aNumber of intervention groups in the studies. Some enrolled studies were not included because the value used for subgroup analysis was not reported

where specific patterns of movement are replicated and load specificity occurs [67]. Therefore, an excess load may limit the development of the SSC, and—to a lesser degree—the coupling of the H reflex, despite the athlete's attempts to maintain the movement pattern. On the other hand, the muscle has been observed to work quasi-isometrically when sprinting [68, 69], allowing for greater tendon lengthening as load intensity increases [70] and for the tendon to act as a power amplifier as it recoils at high velocities [71, 72]. Therefore, if we take into account that the tendon demands are dependent on the movement velocity, the adaptations induced will also be velocity dependent. This appears to also explain why RST is much more effective in the acceleration

phase than in the maximum-velocity phase, where the movement velocity, the SSC, and H reflex are critical [67]. That is, in the acceleration phase, CT is greater than during the maximum-velocity phase, as is the involvement of horizontal force and the larger H:V force ratio. Moreover, it has been suggested that increases in maximal strength are more likely to increase short-distance (5-m) sprint performance [73].

One of the most analyzed variables that has consequently generated the greatest discrepancy among the scientific community in recent years, is the load that should be used when performing sled towing. Some authors [40] have shown that the maximum power production when using these methods occurs with loads around 20% BM, whereas others have

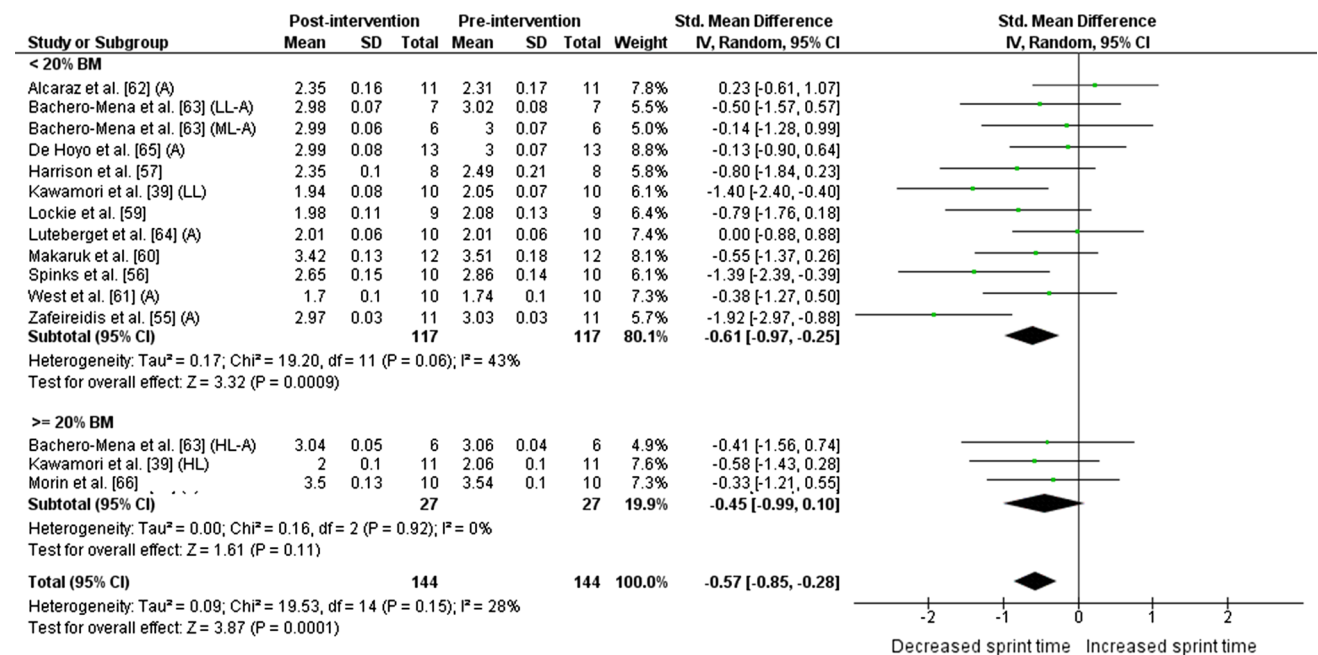


Fig. 4 SMD between post and pre-intervention for sprint time in the acceleration phase based on different load (subgroup analysis). Squares represent the SMD for each trial. Diamonds represent the pooled SMD across trials. A acceleration phase, HL high load, ML

moderate load, LL low load, BM body mass, CI confidence interval, IV independent variable, SD standard deviation, SMD standardized mean difference

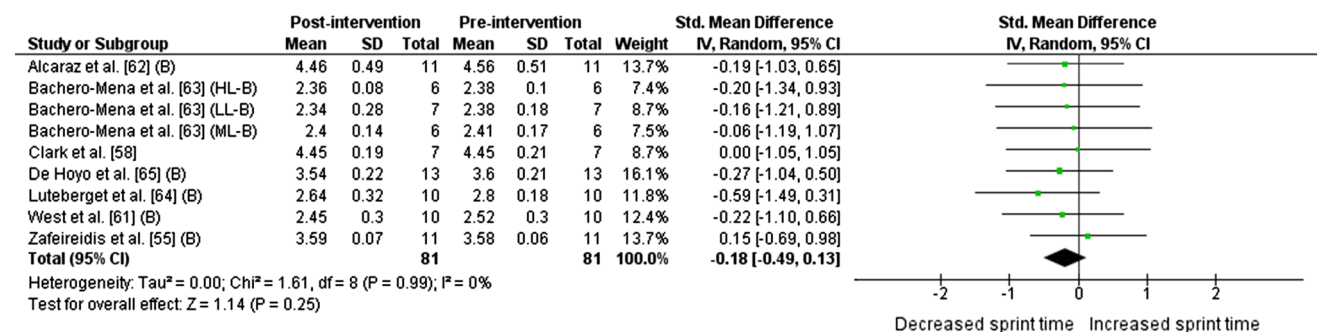


Fig. 5 SMD between post and pre-intervention for sprint time in the maximum-velocity phase. Squares represent the SMD for each trial. Diamonds represent the pooled SMD across trials. B maximum-

velocity phase, CI confidence interval, HL high load, IV independent variable, LL low load, ML moderate load, SD standard deviation, SMD standardized mean difference

suggested that the maximum power output can be achieved with 69–96% BM [41]. It should be noted that Monte et al. [40] calculated and examined the load that maximizes power output with sprinters and for the full sprint (20 m) and analyzed the effect on the kinematics of the joints and segments. However, Cross et al. [41] calculated the maximum power output at the moment when participants (team-sports athletes) reached maximum sprint speed and did not calculate effect on sprint technique. In any case, it is not clear for this training method that training with the load that produces the maximum power output is the most effective method

for improving sprint performance. Hence, more research is needed in this regard.

As mentioned, the present meta-analysis observed that the effect on performance improvement was moderate when comparing loads < 20% BM and loads ≥ 20% BM, with no significant improvements achieved with loads ≥ 20% BM. It should be noted that only three groups [39, 63, 66] used loads ≥ 20% BM and that these three groups were composed of team-sports athletes. Therefore, it cannot be inferred that the use of high loads is more effective than the use of low loads. In fact, the only study to use very high loads (80%

Table 4 Subgroup analysis assessing potential moderating factors for sprint time in the maximum-velocity phase in the studies included in the meta-analysis

Population characteristics	Studies		SMD (95% CI)	Resisted sled training			
	Number ^a	References		ES	<i>I</i> ² (%)	<i>p</i>	<i>p</i> _{Difference}
Age (~ 17 to 27 years)							
< 21 y.o.	7	Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Clark et al. [58] de Hoyo et al. [65]: B Luteberget et al. [64]: B Zafeiridis et al. [55]: B	− 0.17 (− 0.53 to 0.19)	0.26	0	0.35	0.92
≥ 21 y.o.	2	Alcaraz et al. [62]: B West et al. [61]: B	− 0.21 (− 0.81 to 0.40)	0.28	0	0.50	
Sex							
Female ≥ 45% simple	2	Alcaraz et al. [62]: B Luteberget et al. [64]: B	− 0.38 (− 0.99 to 0.24)	0.18	0	0.23	0.47
Male	7	Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Clark et al. [58] de Hoyo et al. [65]: B West et al. [61]: B Zafeiridis et al. [55]: B	− 0.11 (− 0.47 to 0.25)	0.23	0	0.54	
Level							
Recreationally active	4	Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Zafeiridis et al. [55]: B	− 0.04 (− 0.54 to 0.47)	0.21	0	0.89	0.67
Trained	1	Clark et al. [58]	0.00 (− 1.05 to 1.05)	0.00	NA	1	
Highly trained	4	Alcaraz et al. [62]: B de Hoyo et al. [65]: B Luteberget et al. [64]: B West et al. [61]: B	− 0.31 (− 0.73 to 0.11)	0.39	0	0.15	
Sport characteristics							
Team sports	4	Clark et al. [58] de Hoyo et al. [65]: B Luteberget et al. [64]: B West et al. [61]: B	− 0.29 (− 0.73 to 0.15)	0.34	0	0.20	0.84
Individual sports	1	Alcaraz et al. [62]: B	− 0.19 (− 1.03 to 0.65)	0.20	NA	0.65	
Exercise characteristics							
Load (5–20% BM)							
< 20% BM	8	Alcaraz et al. [62]: B Bachero-Mena et al. [63]: LL-B, ML-B Clark et al. [58] de Hoyo et al. [65]: B Luteberget et al. [64]: B West et al. [61]: B Zafeiridis et al. [55]: B	− 0.18 (− 0.50 to 0.14)	0.26	0	0.28	0.97
≥ 20% BM	1	Bachero-Mena et al. [63]: HL-B	− 0.20 (− 1.34 to 0.93)	0.28	NA	0.73	
Frequency (1–3 week ^{−1})							
≤ 2 week ^{−1}	8	Alcaraz et al. [62]: B Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Clark et al. [58] de Hoyo et al. [65]: B Luteberget et al. [64]: B West et al. [61]: B	− 0.23 (− 0.57 to 0.10)	0.25	0	0.17	0.41
> 2 week ^{−1}	1	Zafeiridis et al. [55]: B	0.15 (− 0.69 to 0.98)	0.13	NA	0.73	
Duration (4–8 weeks)							
≤ 6 weeks	2	Alcaraz et al. [62]: B West et al. [61]: B	− 0.21 (− 0.81 to 0.40)	0.28	0	0.50	0.92

Table 4 (continued)

Population characteristics	Studies		SMD (95% CI)	Resisted sled training			
	Number ^a	References		ES	I ² (%)	p	p _{Difference}
> 6 weeks	7	Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Clark et al. [58] de Hoyo et al. [65]: B Luteberget et al. [64]: B Zafeiridis et al. [55]: B	-0.17 (-0.53 to 0.19)	0.26	0	0.35	
Session volume (60–400 m)							
≤ 160 m	6	Alcaraz et al. [62]: B Bachero-Mena et al. [63]: LL-B, ML-B, HL-B De Hoyo et al. [65]: B West et al. [61]: B	-0.20 (-0.58 to 0.18)	0.32	0	0.31	0.87
> 160 m	3	Clark et al. [58] Luteberget [64]: B Zafeiridis et al. [55]: B	-0.15 (-0.67 to 0.38)	0.15	0	0.59	
Total training volume (720–6720 m)							
≤ 2680 m	6	Alcaraz et al. [62]: B Bachero-Mena et al. [63]: LL-B, ML-B, HL-B De Hoyo et al. [65]: B West et al. [61]: B	-0.20 (-0.58 to 0.18)	0.32	0	0.31	0.87
> 2680 m	3	Clark et al. [58] Luteberget et al. [64]: B Zafeiridis et al. [55]: B	-0.15 (-0.67 to 0.38)	0.15	0	0.59	
Surface							
Rigid	3	Clark et al. [58] Luteberget et al. [64]: B West et al. [61]: B	-0.30 (-0.84 to 0.24)	0.31	0	0.28	0.80
Track	5	Alcaraz et al. [62]: B Bachero-Mena et al. [63]: LL-B, ML-B, HL-B Zafeiridis et al. [55]: B	-0.08 (-0.51 to 0.36)	0.20	0	0.73	
Grass	1	de Hoyo et al. [65]: B	-0.27 (-1.04 to 0.50)	0.43	NA	0.49	

Subgroup analyses were performed on SMD between post and pre-intervention sprint time. Median values of continuous variables were used as cut-off values for grouping studies. Changes in moderating factors were calculated as post-intervention minus pre-intervention values

A acceleration phase, B maximum-velocity phase, BM body mass, C full sprint, CI confidence interval, ES Cohen's *d* effect size, HL high load, I² heterogeneity, LL low load, ML medium load, NA not available data, *p* test for overall effect, *p_{Difference}* test for subgroup differences, SMD standardized mean difference

^aNumber of intervention groups in the studies. Some enrolled studies were not included because the value used for subgroup analysis was not reported

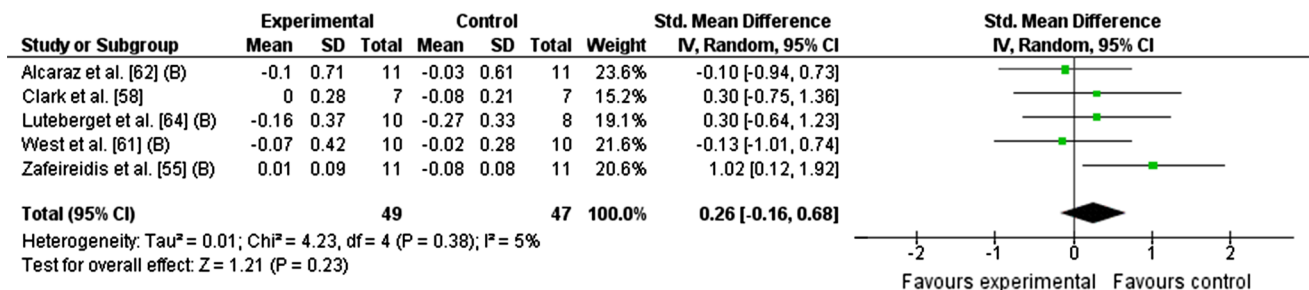


Fig. 6 SMD in post-intervention sprint time between intervention and control athletes for sprint time in the maximum-velocity phase. Squares represent the SMD for each trial. Diamonds represent the

pooled SMD across trials. B maximum-velocity phase, CI confidence interval, IV independent variable, SD standard deviation, SMD standardized mean difference

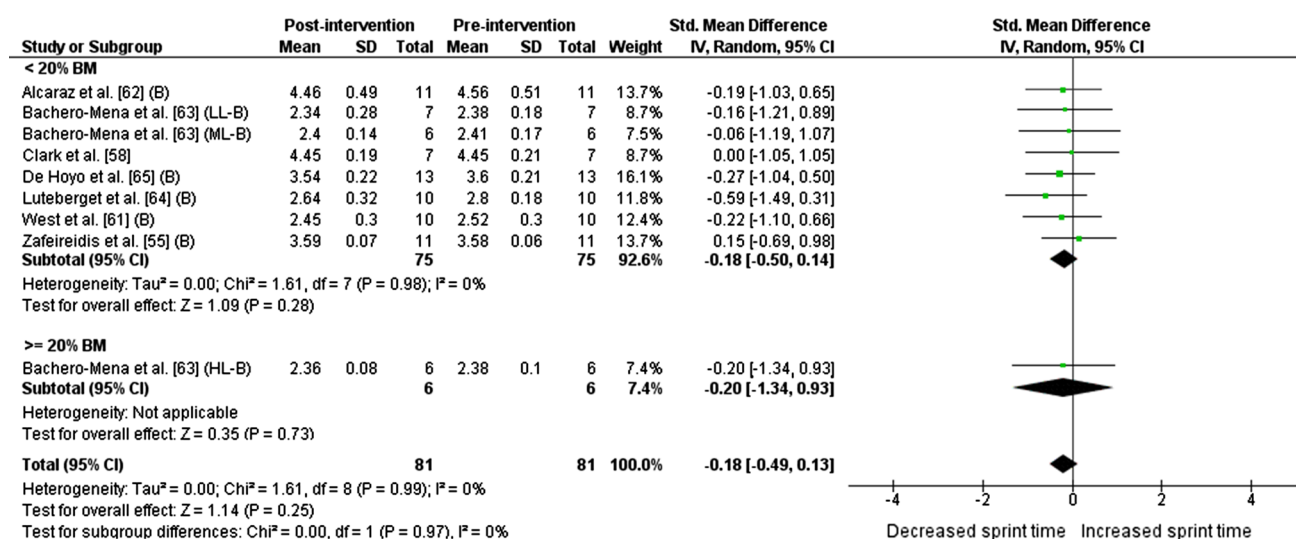


Fig. 7 SMD between post and pre-intervention for sprint time in the maximum-velocity phase based on different load (subgroup analysis). Squares represent the SMD for each trial. Diamonds represent the pooled SMD across trials. *B* maximum-velocity phase, *BM* body

BM) [66] found a small and trivial effect on performance of the 5- and 20-m sprint, respectively, which is similar to those found in the control group with the same training but without overload. Thus, Morin et al. [66] found only a moderate effect on the F_0 (i.e., the force at velocity 0), or what would be equivalent to the maximum theoretical force of the participants, but not on V_0 (i.e., which would be the equivalent of the maximum theoretical velocity of the players), according to the model proposed by the same authors [42]. As a consequence, the proposal of using RST with very heavy loads would only be justified as a tertiary method, which would affect the improvement of the force at low or null velocities “simulating” the movement pattern (and only in athletes who can maintain it). However, it is far from the load and velocity specificity, which are both decisive in the training of maximum power production, as explained by Cormie et al. [28] in their narrative review on maximal power production training. Another problem that can occur when using excessively high loads is that neither the SSC nor the H reflex are properly activated. In fact, training activities aimed at improving SSC performance should fulfill two criteria [15, 74–77]: (1) they should involve skillful, multi-joint movements that transmit forces through the kinetic chain and exploit elastic–reflexive mechanisms; and (2) they should be structured around brief work bouts or clusters separated by frequent rest periods to manage fatigue and emphasize work quality and technique.

Another point to consider is that chronic exposure to movements eliciting the SSC can increase muscle stiffness, which is a potential physiological advantage for sprint ability [78]. An optimal development of mechanical stiffness in the

human leg has a major positive influence on various athletic variables, including RFD, elastic energy storage and utilization, and therefore the SSC, and sprint kinematics (i.e., CT and FT, and SL and frequency) [79]. However, only Alcaraz et al. [62] analyzed the effects of sled towing on stiffness, and they found a slight non-significant reduction after a training period of 4 weeks (with low loads). However, this study observed that the group training without a sled experienced an improvement in stiffness that approached significance, suggesting that training that respects the principle of specificity can have a positive effect on one of the most important variables in sprinting (i.e., stiffness), whereas increasing the load can have the opposite effect. Given these findings and the recommendations based on a mixed-methods approach in which a variety of loads and exercise types are used in a periodized fashion to optimize power output, training with high loads could be an alternative to traditional resistance training. Thereby, using high-load RST to improve maximum strength, athletes train—to some extent—replicating the movement patterns of the sprint. However, when the aim is to work in the area of the force-velocity curve close to the demands of the competition, the RST must be carried out with low loads to develop all the mechanisms involved in this situation, such as the RFD, SSC, H reflex, and leg stiffness.

Regarding the characteristics of the population, statistically significant differences were observed only between levels of participants, with the effect being moderate for both recreationally active (ES 0.75) and trained participants (ES 0.84) but small for highly trained participants (ES 0.30). The explanation for these results lies in the fact that highly

Table 5 Subgroup analysis assessing potential moderating factors for sprint time during full sprint in the studies included in the meta-analysis

Population characteristics	Studies		Resisted sled training				
	Number ^a	References	SMD (95% CI)	ES	I^2 (%)	p	$p_{\text{Difference}}$
Age (~ 17 to 27 years)							
< 21 y.o.	6	Bachero-Mena et al. [63]: LL-C, ML-C, HL-C De Hoyo et al. [65]: C Luteberget et al. [64]: C Zafeiridis et al. [55]: C	−0.41 (−0.80 to −0.02)	0.37	0	0.04	0.83
≥ 21 y.o.	4	Alcaraz et al. [62]: C Makaruk et al. [60] Morin et al. [66] West et al. [61]: C	−0.35 (−0.78 to 0.08)	0.35	0	0.11	
Sex							
Female ≥ 45% sample	3	Alcaraz et al. [62]: C Luteberget et al. [64]: C Makaruk et al. [60]	−0.42 (−0.91 to 0.07)	0.41	0	0.09	0.84
Male	7	Bachero-Mena et al. [63]: LL-C, ML-C, HL-C De Hoyo et al. [65]: C Morin et al. [66] West et al. [61]: C Zafeiridis et al. [55]: C	−0.36 (−0.71 to −0.01)	0.34	0	0.04	
Level							
Recreationally active	4	Bachero-Mena et al. [63]: LL-C, ML-C, HL-C Zafeiridis et al. [55]: C	−0.38 (−0.90 to 0.13)	0.35	0	0.14	0.91
Trained	0	—	—	—	—	—	
Highly trained	4	Alcaraz et al. [62]: C De Hoyo et al. [65]: C Luteberget et al. [64]: C West et al. [61]: C	−0.35 (−0.77 to 0.08)	0.32	0	0.11	
Sport characteristics							
Team sports	5	De Hoyo et al. [65]: C Luteberget et al. [64]: C Makaruk et al. [60] Morin et al. [66] West et al. [61]: C	−0.43 (−0.81 to −0.06)	0.42	0	0.02	0.51
Individual sports	1	Alcaraz et al. [62]: C	−0.12 (−0.96 to 0.71)	0.12	NA	0.77	
Exercise characteristics							
Load (5–80% BM)							
< 20% BM	8	Alcaraz et al. [62]: C Bachero-Mena et al. [63]: LL-C, ML-C De Hoyo et al. [65]: C Luteberget et al. [64]: C Makaruk et al. [60] West et al. [61]: C Zafeiridis et al. [55]: C	−0.38 (−0.70 to −0.07)	0.35	0	0.02	0.99
≥ 20% BM	2	Bachero-Mena et al. [63]: HL-C Morin et al. [66]	−0.39 (−1.09 to 0.32)	0.42	0	0.28	
Frequency (1–3 week ^{−1})							
≤ 2 week ^{−1}	9	Alcaraz et al. [62]: C Bachero-Mena et al. [63]: LL-C, ML-C, HL-C de Hoyo et al. [65]: C Luteberget et al. [64]: C Makaruk et al. [60] Morin et al. [66] West et al. [61]: C	−0.37 (−0.67 to −0.06)	0.35	0	0.02	0.76
> 2 week ^{−1}	1	Zafeiridis et al. [55]: C	−0.51 (−1.36 to 0.35)	0.46	NA	0.24	

Table 5 (continued)

Population characteristics	Studies		Resisted sled training				
	Number ^a	References	SMD (95% CI)	ES	I^2 (%)	p	$p_{\text{Difference}}$
Duration (4–10 weeks)							
≤ 6 weeks	2	Alcaraz et al. [62]: C West et al. [61]: C	− 0.24 (− 0.85 to 0.37)	0.24	0	0.44	0.61
> 6 weeks	8	Bachero-Mena et al. [63]: LL-C, ML-C, HL-C De Hoyo et al. [65]: C Luteberget et al. [64]: C Makaruk et al. [60] Morin et al. [66] Zafeiridis et al. [55]: C	− 0.42 (− 0.75 to − 0.10)	0.39	0	0.01	
Session volume (60–360 m)							
≤ 160 m	7	Alcaraz et al. [62]: C Bachero-Mena et al. [63]: LL-C, ML-C, HL-C De Hoyo et al. [65]: C Morin et al. [66] West et al. [61]: C	− 0.29 (− 0.65 to 0.06)	0.29	0	0.10	0.40
> 160 m	3	Luteberget et al. [64]: C Makaruk et al. [60] Zafeiridis et al. [55]: C	− 0.56 (− 1.05 to − 0.06)	0.53	0	0.03	
Total training volume (720–6720 m)							
≤ 2680 m	7	Alcaraz et al. [62]: C Bachero-Mena et al. [63]: LL-C, ML-C, HL-C De Hoyo et al. [65]: C Morin et al. [66] West et al. [61]: C	− 0.29 (− 0.65 to 0.06)	0.29	0	0.10	0.40
> 2680 m	3	Luteberget et al. [64]: C Makaruk et al. [60] Zafeiridis et al. [55]: C	− 0.56 (− 1.05 to − 0.06)	0.53	0	0.03	
Surface							
Rigid	2	Luteberget et al. [64]: C West et al. [61]: C	− 0.49 (− 1.12 to 0.14)	0.42	0	0.13	0.90
Track	5	Alcaraz et al. [62]: C Bachero-Mena et al. [63]: LL-C, ML-C, HL-C Zafeiridis et al. [55]: C	− 0.31 (− 0.75 to 0.12)	0.31	0	0.16	
Grass	3	De Hoyo et al. [65]: C Makaruk et al. [60] Morin et al. [66]	− 0.40 (− 0.88 to 0.07)	0.42	0	0.10	

Subgroup analyses were performed on SMD between post and pre-intervention sprint time. Median values of continuous variables were used as cut-off values for grouping studies. Changes in moderating factors were calculated as post-intervention minus pre-intervention values

A acceleration phase, B maximum-velocity phase, BM body mass, C full sprint, CI confidence interval, ES Cohen's d effect size, HL high load, I^2 heterogeneity, LL low load, ML medium load, NA not available data, p test for overall effect, $p_{\text{Difference}}$ test for subgroup differences, SMD standardized mean difference

^aNumber of intervention groups in the studies. Some enrolled studies were not included because the value used for subgroup analysis was not reported

trained athletes have a lower margin of improvement than recreationally active or trained athletes when any training protocol is applied [80]. On the other hand, it has been observed that, when an RST is applied to improve acceleration capacity, the effect is moderate in team sports (ES 0.66) but small in individual sports (athletics) (ES 0.23). It is necessary to consider, in this case, that only one study was included [60] that analyzed the effects of RST on national level sprinters and jumpers. Thus, more studies are needed

in this type of population, since RST is used regularly in the design of training plans in athletics, specifically in force-velocity disciplines (such as sprints, jumps, and hurdles), and understanding is needed on the effects of RST on sprint performance in both trained and highly trained athletes for proper programming during the season. The effect shown for team sports is also statistically significant. Therefore, the RST is clearly recommended for the improvement of early acceleration in sports such as soccer or rugby, particularly

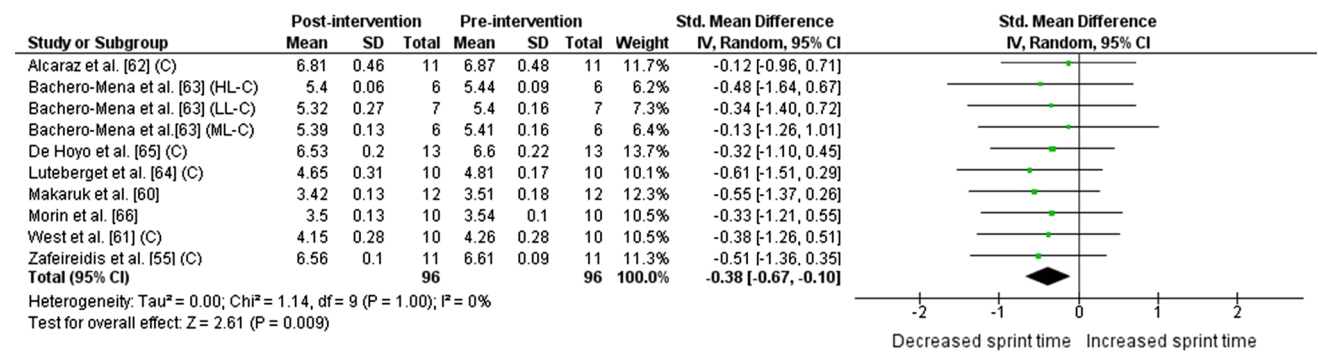


Fig. 8 SMD between post and pre-intervention for sprint time during full sprint. Squares represent the SMD for each trial. Diamonds represent the pooled SMD across trials. *C* full sprint, *CI* confidence inter-

val, *HL* high load, *IV* independent variable, *LL* low load, *ML* moderate load, *SD* standard deviation, *SMD* standardized mean difference

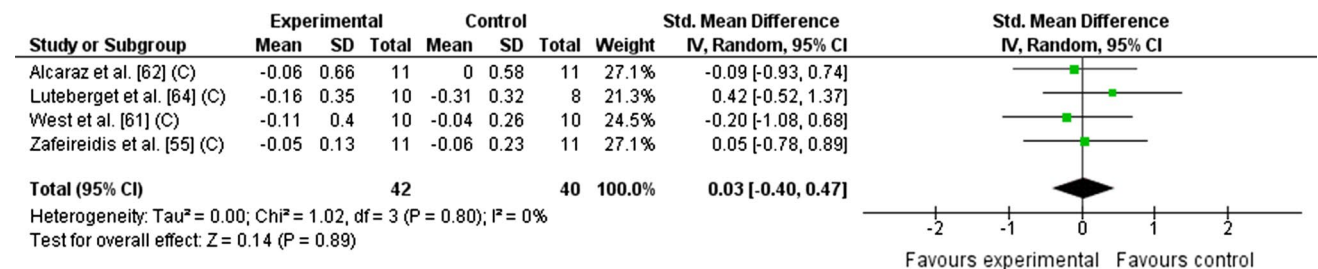


Fig. 9 SMD in post-intervention sprint time between intervention and control athletes for sprint time during full sprint. Squares represent the SMD for each trial. Diamonds represent the pooled SMD across

trials. *C* full sprint, *CI* confidence interval, *IV* independent variable, *SD* standard deviation, *SMD* standardized mean difference

since sprinting is the most frequent physical action for scoring and assisting players before goals [1], and total sprint distance and number of sprints undertaken during games have increased significantly in different European league players in the last decade [81].

Regarding the characteristics of training, weekly training frequencies > 2 days were observed to produce a significantly greater effect than frequencies ≤ 2 (ES 1.85 vs. 0.52). However, these data must be taken with caution since only one study was included [53] that had applied > 2 days (3 days·week⁻¹) of training. In addition, this study was conducted with physical education students, who had a lower level of sprint performance and, a priori, higher potential for improvement than those who are highly trained, as noted in Sect. 3.3. Therefore, training frequencies between 2–3 days·week⁻¹ can be optimal for the development of acceleration capacity when using RST. With respect to the duration of the program, four groups [57, 59, 61, 62] had a duration of 4–6 weeks, with a small and non-significant effect, whereas 11 groups [39, 55, 56, 60, 63–66] used durations of 6–10 weeks, with a significant ($p < 0.001$) moderate effect (ES 0.63). The explanation for this finding may be that shorter durations do not produce sufficient neuromuscular and mechanical adaptations to have a positive effect on sprint performance [82]. Consequently, when designing a

program for the development of sprint performance through RST, based on the results of the present meta-analysis, we recommend a minimum duration of > 6 weeks.

Another variable that should be considered when designing training programs with RST is volume, both for each session and for the microcycle. The volume per session varied widely, from 60 to 360 m. Although most (ten) groups included volumes < 160 m, volumes between 160 and 360 m (five groups) produced greater effects than lower volumes (ES 0.92 vs. 0.46, respectively). Similarly, when weekly volumes were compared, higher volumes (> 2680 m) produced a moderate effect (ES 0.83) compared with volumes 720–2680 m (ES 0.53), although the effects were statistically significant for both volume per session and weekly volume. Therefore, we can infer that slightly higher volumes, both per session and microcycle, have a greater effect; however, whether volume is as important as other variables that have been previously analyzed is unclear.

Finally, with respect to the early acceleration phase, it has been observed that the effect differs according to the surface used. For example, rigid surfaces or athletic tracks have a moderate effect (ES 0.69 and 0.64, respectively) and grass produces a small effect. These results can be explained by the high variability in friction related to the different types of surfaces (natural vs. artificial turf). Since the coefficient of

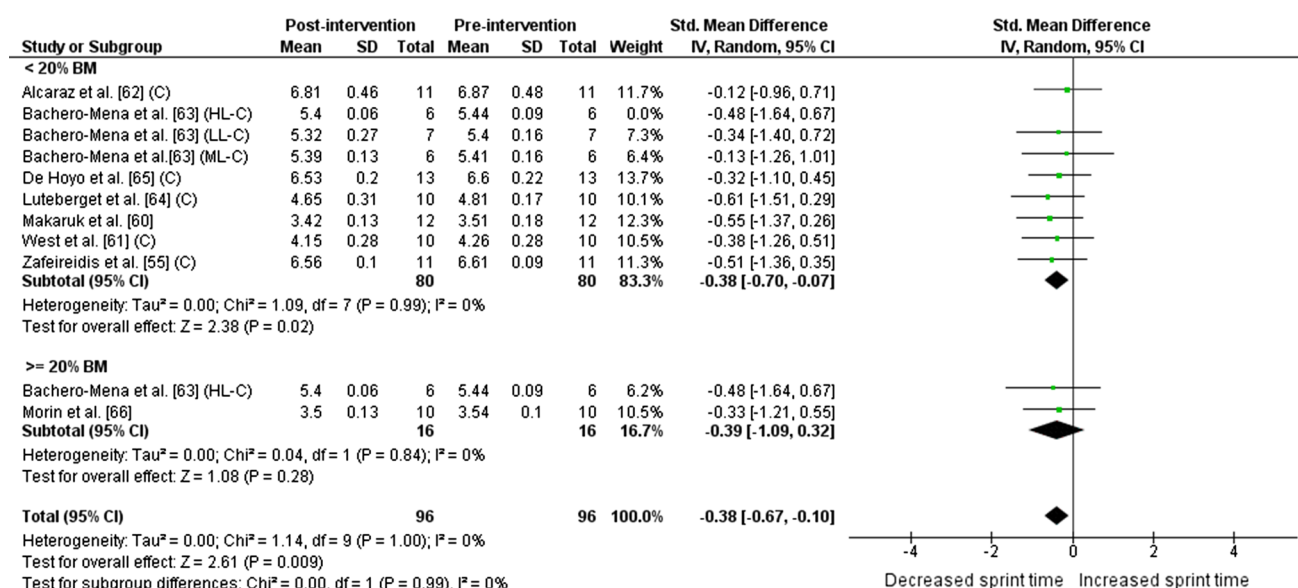


Fig. 10 SMD between post and pre-intervention for sprint time during full sprint based on different load (subgroup analysis). Squares represent the SMD for each trial. Diamonds represent the pooled

friction differs greatly between surfaces, this largely affects resistance when using RST, as suggested by Linthorne and Cooper [46]. These authors concluded that different sprint surfaces would elicit varying degrees of coefficient of friction.

4.2 Maximum-Velocity Phase and Full Sprint

Unlike the findings for the acceleration phase, RST has a small effect on performance in the maximum-velocity phase (ES 0.27) and/or when performing sprints ≥ 20 m (ES 0.36); however, in the latter case, the effect was significant. These findings can be explained by the fact that resistance coming from the friction between the surface of the sled and the contact surface (track, grass, etc.) when the sprint is performed with a sled will be different if the sled is stopped or in motion because of the inertia of the system. Therefore, when the maximum power output is calculated, the maximum power output in sprinting usually occurs in the first steps, and the horizontal force and power are decreased by 82% and 63%, respectively, from the first to the last step when using 20-m RST [40]. Therefore, if the aim is to develop maximum power production in the maximum-velocity phase, training strategies other than sled towing may be needed, since vertical forces are predominant in this phase [9, 16, 17]. Perhaps plyometric training or another type of resisted training, such as the weighted vest, could produce a greater effect in these phases.

When analyzing both the characteristics of the population and the training method for the maximum-velocity phase, the effects were small or trivial and not significant in all

SMD across trials. *BM* body mass, *C* full sprint, *CI* confidence interval, *HL* high load, *IV* independent variable, *LL* low load, *ML* moderate load, *SD* standard deviation, *SMD* standardized mean difference

cases. However, sub-analysis for the full sprint (≥ 20 m) indicated that, although the effects were still trivial or small in some cases, some were significant. For example, regarding the population characteristics, the effect was small but significant for age < 21 years (ES 0.37; $p = 0.04$), men (ES 0.34; $p = 0.04$), and team sports (ES 0.42; $p = 0.02$). With respect to the training characteristics, the effect was small in all cases and significant with loads $< 20\%$ BM (ES 0.35; $p = 0.02$), weekly training frequencies of ≤ 2 days (ES 0.35; $p = 0.02$), for duration > 6 weeks (ES 0.39; $p = 0.01$), volumes per session > 160 m (ES 0.53; $p = 0.03$), and weekly values > 2680 m (ES 0.53; $p = 0.03$).

Although this meta-analysis answers many of the questions from the scientific literature, there is still a significant lack of research that focuses on highly trained athletes in individual sports such as athletics, in both males and females of different ages.

5 Conclusions

RST has been used extensively in both team and individual sports to improve sprint performance. However, to date, there has been no consensus on whether this training method actually improves sprint performance, in either the acceleration or the maximum-velocity phases. Furthermore, whether the effects differ according to population characteristics is unclear, as is the optimal training load (intensity, volume, etc.) for adaptations to be optimised. Based on the present systematic review with meta-analysis, it can be affirmed that RST is an effective method to improve sprint performance,

mainly via improvement of the early acceleration phase. However, it cannot be said that this method is more effective than the same training without overload. Regarding the population characteristics, the effect is greater in men, recreationally active or trained, but is small in highly trained individuals who practice team sports such as football (soccer) or rugby. Finally, with regards to the training characteristics, the intensity (load) is not a determinant of sprint performance improvement, but the recommended volume is > 160 m per session, and approximately 2680 m per week, with a training frequency of 2–3 times per week, for at least 6 weeks. Finally, rigid surfaces appear to enhance the effect of RST on sprint performance.

Based on these findings and given the limitations of this meta-analysis, we provide the following answers to the questions posed by coaches and the scientific community in Sect. 1:

1. Does RST improve sprint performance? Yes, but the improvements will depend on the training phase.
2. Will the effect be greater when RST is used for the different phases of sprint? Yes. For the early acceleration phase (≤ 10 m), there will be a reduction in the average sprint time of 2.3%; for the maximum-velocity phase (≥ 15 -m flying sprints at maximum intensity using a run-in distance of ≥ 10 m), the time will be reduced by 1.7%; and for the full sprint (≥ 20 m), the reduction will be 1.5%. The effect is moderate and significant only for the early acceleration phase.
3. Is RST more effective than UR sprinting? No; no difference was observed between groups with sleds and controls (same training without sled). Therefore, RST or UR will produce a similar level of adaptation.
4. What is the optimal load to use when applying RST for obtaining higher sprint adaptations? There is no optimal load for RST, since it will depend on the desired objective. However, when training with sled towing as a secondary method (i.e., replicating sprint demands in terms of movement pattern, load, and movement velocity) but with a slight overload, the load must never be $> 20\%$ BM. Conversely, when the aim is to improve maximum strength, “respecting” the movement pattern, loads $> 20\%$ BM could be suitable, as long as the athlete does not substantially modify his/her running technique. Loads $> 20\%$ BM should not be used with low-level or inexperienced athletes with sled towing.
5. Should the load be different for the different phases of sprint? Yes, keeping in mind that the main effect of RST occurs in the early acceleration phase. For the maximum-velocity phase, instead of using a different load, perhaps another sprint training method with a more vertical resistance component, like weighted vests, could be used.

6. Does RST have a different effect on athletes depending on age? This is not clear. Therefore, more studies comparing the effects of RST on different age groups are necessary to determine whether the effect will differ.
7. Will the effect be greater if the athlete is highly trained? No, in fact, for the acceleration phase, the effect is significantly greater in recreationally active and trained athletes than in highly trained athletes.
8. How many days per week should RST be applied? These methods should be applied between 2 and 3 days per week, depending on the sprint demands of the sport in which the athlete is working.
9. How many weeks should RST be applied? The effect is significantly greater when the training is > 6 weeks.
10. Can the surface affect the performance adaptations? Yes, the effect is greater for rigid surfaces than for grass, probably due to the lower friction that exists with the same load on this surface.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of interest Pedro E. Alcaraz, Jorge Carlos-Vivas, Bruno O. Oponjuru, and Alejandro Martínez-Rodríguez have no conflicts of interest relevant to the content of this review.

References

1. Faude O, Koch T, Meyer T. Straight sprinting is the most frequent action in goal situations in professional football. *J Sports Sci.* 2012;30(7):625–31.
2. Haugen TA, Tønnessen E, Seiler S. Speed and countermovement-jump characteristics of elite female soccer players, 1995–2010. *Int J Sports Physiol Perform.* 2012;7(4):340–9.
3. Haugen TA, Tønnessen E, Seiler S. Anaerobic performance testing of professional soccer players 1995–2010. *Int J Sports Physiol Perform.* 2013;8(2):148–56.
4. Vescovi JD. Impact of maximum speed on sprint performance during high-level youth female field hockey matches: Female Athletes in Motion (FAiM) study. *Int J Sports Physiol Perform.* 2014;9(4):621–6.
5. Mann RA, Hagy J. Biomechanics of walking, running, and sprinting. *Am J Sports Med.* 1980;8(5):345–50.
6. Young W, Benton D, Duthie G, Pryor J. Resistance training for short sprints and maximum-speed sprints. *Strength Cond J.* 2001;23(2):7–13.
7. Mero A, Komi P, Gregor R. Biomechanics of sprint running. A review. *Sports Med.* 1992;13(6):376–92.
8. Alexander RM. Mechanics of skeleton and tendons. *Handbook of physiology—the nervous system.* Am Physiol Soc. 1981;2:17–42.
9. Rabita G, Dorel S, Slawinski J, Saez-de-Villarreal E, Couturier A, Samozino P, et al. Sprint mechanics in world-class athletes: a

- new insight into the limits of human locomotion. *Scand J Med Sci Sports*. 2015;25(5):583–94.
10. Mero A, Komi PV. Force-, EMG-, and elasticity-velocity relationships at submaximal, maximal and supramaximal running speeds in sprinters. *Eur J Appl Physiol Occup Physiol*. 1986;55(5):553–61.
 11. Delecluse C. Influence of strength training on sprint running performance. *Sports Med*. 1997;24(3):147–56.
 12. Weyand PG, Sandell RF, Prime DNL, Bundle MW. The biological limits to running speed are imposed from the ground up. *J Appl Physiol*. 2010;108(4):950–61.
 13. Weyand PG, Sternlight DB, Bellizzi MJ, Wright S. Faster top running speeds are achieved with greater ground forces not more rapid leg movements. *J Appl Physiol*. 2000;89(5):1991–9.
 14. Clark KP, Weyand PG. Are running speeds maximized with simple-spring stance mechanics? *J Appl Physiol*. 2014;117(6):604–15.
 15. DeWeese BH, Bellon C, Magrum E, Taber CB, Suchomel TJ. Strengthening the springs: improving sprint performance via strength training. *Track Tech*. 2016;9(3):8–20.
 16. Nagahara R, Mizutani M, Matsuo A, Kanehisa H, Fukunaga T. Step-to-step spatiotemporal variables and ground reaction forces of intra-individual fastest sprinting in a single session. *J Sports Sci*. 2017;36(12):1392–401.
 17. Nagahara R, Mizutani M, Matsuo A, Kanehisa H, Fukunaga T. Association of sprint performance with ground reaction forces during acceleration and maximal speed phases in a single sprint. *J Appl Biomech*. 2017;34:104–10.
 18. Dalleau G, Belli A, Bourdin M, Lacour J-R. The spring-mass model and the energy cost of treadmill running. *Eur J Appl Physiol Occup Physiol*. 1998;77(3):257–63.
 19. Voigt M, Bojsen-Møller F, Simonsen EB, Dyhre-Poulsen P. The influence of tendon Young's modulus, dimensions and instantaneous moment arms on the efficiency of human movement. *J Biomech*. 1995;28(3):281–91.
 20. Plisk SS. Speed, agility, and speed-endurance development. In: Baechle TR, Earle RW, editors. *National strength and conditioning association: essentials of strength training & conditioning*. 2nd ed. Champaign IL: Human Kinetics; 2000. p. 471–91.
 21. MacDougall D, Sale D. Continuous vs. interval training: a review for the athlete and the coach. *Can J Appl Sport Sci*. 1981;6(2):93–7.
 22. Iacono AD, Martone D, Milic M, Padulo J. Vertical-vs. horizontal-oriented drop jump training: Chronic effects on explosive performances of elite handball players. *J Strength Cond Res*. 2017;31(4):921–31.
 23. Rumpf MC, Lockie RG, Cronin JB, Jalilvand F. Effect of different sprint training methods on sprint performance over various distances: a brief review. *J Strength Cond Res*. 2016;30(6):1767–85.
 24. Freitas TT, Martinez-Rodriguez A, Calleja-González J, Alcaraz PE. Short-term adaptations following complex training in team-sports: a meta-analysis. *PLoS One*. 2017;12(6):e0180223.
 25. Seitz LB, Reyes A, Tran TT, de Villarreal ES, Haff GG. Increases in lower-body strength transfer positively to sprint performance: a systematic review with meta-analysis. *Sports Med*. 2014;44(12):1693–702.
 26. Hill A. The heat of shortening and the dynamic constants of muscle. *Proc R Soc Lond B Biol Sci*. 1938;126(843):136–95.
 27. Hill A, Long C, Lupton H. The effect of fatigue on the relation between work and speed, in contraction of human arm muscles. *J Physiol*. 1924;58(4–5):334–7.
 28. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power part 2-training considerations for improving maximal power production. *Sports Med*. 2011;41(2):125–46.
 29. Randell AD, Cronin JB, Keogh JWL, Gill ND. Transference of strength and power adaptation to sports performance-horizontal and vertical force production. *Strength Cond J*. 2010;32(4):100–6.
 30. Haff GG, Nimphius S. Training principles for power. *Strength Cond J*. 2012;34(6):2–12.
 31. Alcaraz PE, Palao JM, Elvira JLL, Linthorne NP. Effects of three types of resisted sprint training devices on the kinematics of sprinting at maximum velocity. *J Strength Cond Res*. 2008;22(3):890–7.
 32. Kawamori N, Newton R, Nosaka K. Effects of weighted sled towing on ground reaction force during the acceleration phase of sprint running. *J Sports Sci*. 2014;32(12):1139–45.
 33. Brown SC, Craik FI. Encoding and retrieval of information. In: Tulving E, Craik FIM, editors. *The Oxford handbook of memory*. 2000. p. 93–107.
 34. Haff GG, Stone MH. Methods of developing power with special reference to football players. *Strength Cond J*. 2015;37(6):2–16.
 35. Alcaraz PE, Palao JM, Elvira JL. Determining the optimal load for resisted sprint training with sled towing. *J Strength Cond Res*. 2009;23(2):480–5.
 36. Jakalski K. The pros and cons of using resisted and assisted training methods with high school sprinters: parachutes, tubing and towing. *Track Coach*. 1998;144:4585–9.
 37. Lockie RG, Murphy AJ, Spinks CD. Effects of resisted sled towing on sprint kinematics in field-sport athletes. *J Strength Cond Res*. 2003;17(4):760–7.
 38. Cottle CA, Carlson LA, Lawrence MA. Effects of sled towing on sprint starts. *J Strength Cond Res*. 2014;28(5):1241–5.
 39. Kawamori N, Newton RU, Hori N, Nosaka K. Effects of weighted sled towing with heavy versus light load on sprint acceleration ability. *J Strength Cond Res*. 2014;28(10):2738–45.
 40. Monte A, Nardello F, Zamparo P. Sled towing: the optimal overload for peak power production. *Int J Sports Physiol Perform*. 2016;12(8):1052–8.
 41. Cross MR, Brughelli M, Samozino P, Brown SR, Morin J-B. Optimal loading for maximising power during sled-resisted sprinting. *Int J Sports Physiol Perform*. 2017;12(8):1069–77.
 42. Samozino P, Rabita G, Dorel S, Slawinski J, Peyrot N, Saez de Villarreal E, et al. A simple method for measuring power, force, velocity properties, and mechanical effectiveness in sprint running. *Scand J Med Sci Sports*. 2016;26(6):648–58.
 43. Cronin JB, Hansen K, Kawamori N, McNair P. Effects of weighted vests and sled towing on sprint kinematics. *Sports Biomech*. 2008;7(2):160–72.
 44. Maulder PS, Bradshaw EJ, Keogh JWL. Kinematic alterations due to different loading schemes in early acceleration sprint performance from starting blocks. *J Strength Cond Res*. 2008;22(6):1992–2002.
 45. Martinez-Valencia MA, Romero-Arenas S, Elvira JLL, Gonzalez-Rave JM, Navarro-Valdivielso F, Alcaraz PE. Effects of sled towing on peak force, the rate of force development and sprint performance during the acceleration phase. *J Hum Kinet*. 2015;46(1):139–48.
 46. Linthorne NP, Cooper JE. Effect of the coefficient of friction of a running surface on sprint time in a sled-towing exercise. *Sports Biomech*. 2013;12(2):175–85.
 47. Petrakos G, Morin JB, Egan B. Resisted sled sprint training to improve sprint performance: A systematic review. *Sports Med*. 2016;46(3):381–400.
 48. Moher D, Liberati A, Tetzlaff J, Altman DG, Prisma G. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009;6(7):e1000097.
 49. Bhogal SK, Teasell RW, Foley NC, Speechley MR. The PEDro scale provides a more comprehensive measure of methodological quality than the Jadad scale in stroke rehabilitation literature. *J Clin Epidemiol*. 2005;58(7):668–73.

50. de Morton NA. The PEDro scale is a valid measure of the methodological quality of clinical trials: a demographic study. *Aust J Physiother.* 2009;55(2):129–33.
51. Maher CG, Sherrington C, Herbert RD, Moseley AM, Elkins M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Phys Ther.* 2003;83(8):713–21.
52. Cohen J. Statistical power analysis for the behavioral sciences (revised ed.). New York: Academic; 1977.
53. Hopkins W, Marshall S, Batterham A, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc.* 2009;41(1):3.
54. Higgins JP, Green S. *Cochrane handbook for systematic reviews of interventions.* New York: Wiley; 2011.
55. Zafeiridis A, Saraslanidis P, Manou V, Ioakimidis P, Dipla K, Kellis S. The effects of resisted sled-pulling sprint training on acceleration and maximum speed performance. *J Sports Med Phys Fit.* 2005;45(3):284–90.
56. Spinks CD, Murphy AJ, Spinks WL, Lockie RG. The effects of resisted sprint training on acceleration performance and kinematics in soccer, rugby union, and Australian football players. *J Strength Cond Res.* 2007;21(1):77–85.
57. Harrison AJ, Bourke G. The effect of resisted sprint training on speed and strength performance in male rugby players. *J Strength Cond Res.* 2009;23(1):275–83.
58. Clark KP, Stearne DJ, Walts CT, Miller AD. The longitudinal effects of resisted sprint training using weighted sleds vs weighted vests. *J Strength Cond Res.* 2010;24(12):3287–95.
59. Lockie RG, Murphy AJ, Schultz AB, Knight TJ, de Jonge XAKJ. The effects of different speed training protocols on sprint acceleration kinematics and muscle strength and power in field sport athletes. *J Strength Cond Res.* 2012;26(6):1539–50.
60. Makaruk B, Sozański H, Makaruk H, Sacewicz T. The effects of resisted sprint training on speed performance in women. *Hum Mov Sci.* 2013;14(2):116–22.
61. West DJ, Cunningham DJ, Bracken RM, Bevan HR, Crewther BT, Cook CJ, et al. Effects of resisted training on acceleration in professional rugby union players. *J Strength Cond Res.* 2013;27(4):1014–8.
62. Alcaraz PE, Elvira JLL, Palao JM. Kinematic, strength, and stiffness adaptations after a short-term sled towing training in athletes. *Scand J Med Sci Sports.* 2014;24(2):279–90.
63. Bachero-Mena B, Gonzalez-Badillo JJ. Effects of resisted sprint training on acceleration with three different loads accounting for 5, 12.5, and 20% of body mass. *J Strength Cond Res.* 2014;28(10):2954–60.
64. Luteberget LS, Raastad T, Seynnes O, Spencer M. Effect of traditional and resisted sprint training in highly trained female team handball players. *Int J Sports Physiol Perform.* 2015;10(5):642–7.
65. de Hoyo M, Gonzalo-Skok O, Sañudo B, Carrascal C, Plaza-Armas JR, Camacho-Candil F, et al. Comparative effects of in-season full-back squat, resisted sprint training, and plyometric training on explosive performance in U-19 elite soccer players. *J Strength Cond Res.* 2016;30(2):368–77.
66. Morin JB, Petrakos G, Jimenez-Reyes P, Brown SR, Samozino P, Cross MR. Very-heavy sled training for improving horizontal force output in soccer players. *Int J Sports Physiol Perform.* 2016;12(6):840–4.
67. Ross A, Leveritt M, Riek S. Neural influences on sprint running. *Sports Med.* 2001;31(6):409–25.
68. Van Hooren B, Bosch F. Is there really an eccentric action of the hamstrings during the swing phase of high-speed running? part I: a critical review of the literature. *J Sports Sci.* 2017;35(23):2313–21.
69. Van Hooren B, Bosch F. Is there really an eccentric action of the hamstrings during the swing phase of high-speed running? Part II: implications for exercise. *J Sports Sci.* 2017;35(23):2322–33.
70. Reeves ND, Narici MV. Behavior of human muscle fascicles during shortening and lengthening contractions in vivo. *J Appl Physiol.* 2003;95(3):1090–6.
71. Ettema G. Mechanical efficiency and efficiency of storage and release of series elastic energy in skeletal muscle during stretch-shorten cycles. *J Exp Biol.* 1996;199(9):1983–97.
72. Nagano A, Komura T, Fukushima S. Effects of the length ratio between the contractile element and the series elastic element on an explosive muscular performance. *J Electromyogr Kinesiol.* 2004;14(2):197–203.
73. Comfort P, Bullock N, Pearson SJ. A comparison of maximal squat strength and 5-, 10-, and 20-meter sprint times, in athletes and recreationally trained men. *J Strength Cond Res.* 2012;26(4):937–40.
74. McLean B. Biomechanics of running. In: Hawley JA, editor. *Handbook of sports medicine and science: running.* John Wiley & Sons; 2008. p. 28–43.
75. Schmidtbleicher D. Strength training: part 2: structural analysis of motor strength qualities and its application to training. *Sci Period Res Tech Sport.* 1985;4:1–10.
76. Schmidtbleicher D. Strength training: part 1: structural analysis of motor strength qualities and its application to training. *Sci Period Res Tech Sport.* 1985;4:1–12.
77. Siff M, Verkhoshansky Y. *Supertraining.* Denver: Supertraining Institute; 2003.
78. Farley CT, Gonzalez O. Leg stiffness and stride frequency in human running. *J Biomech.* 1996;29(2):181–6.
79. Brughelli M, Cronin J. A review of research on the mechanical stiffness in running and jumping: methodology and implications. *Scand J Med Sci Sports.* 2008;18(4):417–26.
80. Kraemer WJ, Adams K, Cafarelli E, Dudley GA, Dooly C, Feigenbaum MS, American College of Sports Medicine position stand, et al. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc.* 2002;34(2):364–80.
81. Barnes C, Archer D, Hogg B, Bush M, Bradley P. The evolution of physical and technical performance parameters in the English Premier League. *Int J Sports Med.* 2014;35(13):1095–100.
82. Bolger R, Lyons M, Harrison AJ, Kenny IC. Sprinting performance and resistance-based training interventions: a systematic review. *J Strength Cond Res.* 2015;29(4):1146–56.

Affiliations

Pedro E. Alcaraz^{1,2} · Jorge Carlos-Vivas¹ · Bruno O. Oponjuru¹ · Alejandro Martínez-Rodríguez³

¹ UCAM Research Center for High Performance Sport, Catholic University of Murcia, Murcia, Spain

² Faculty of Sport Sciences, UCAM, Catholic University of Murcia, Murcia, Spain

³ Department of Analytical Chemistry, Nutrition and Food Sciences, Faculty of Sciences, University of Alicante, Alicante, Spain